

Lesson 12 Combinatorial Proofs

1. Write the 10th row of Pascal Triangle.
2. What is the coefficient of x^2 in $(2x - 1)^4$?
3. Use the binomial theorem to show that $9^n = \sum_{k=0}^n (-1)^k \binom{n}{k} 10^{n-k}$.
4. Let $n \in \mathbb{N}$. Give a combinatorial proof for the following identity:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots$$

(Optional) Let $n \in \mathbb{N}$. Give a combinatorial proof for the following identity:

$$\sum_{k=0}^n \binom{n}{k} 2^{n-k} = 3^n$$

Hint: Consider ternary strings (Lesson 11 Exercises #8f).